Golden: Lightweight Non-Interactive Distributed Key Generation

Benedikt Bünz

NYU,
Espresso Systems

Kevin Choi

Seoul National University,
NYU

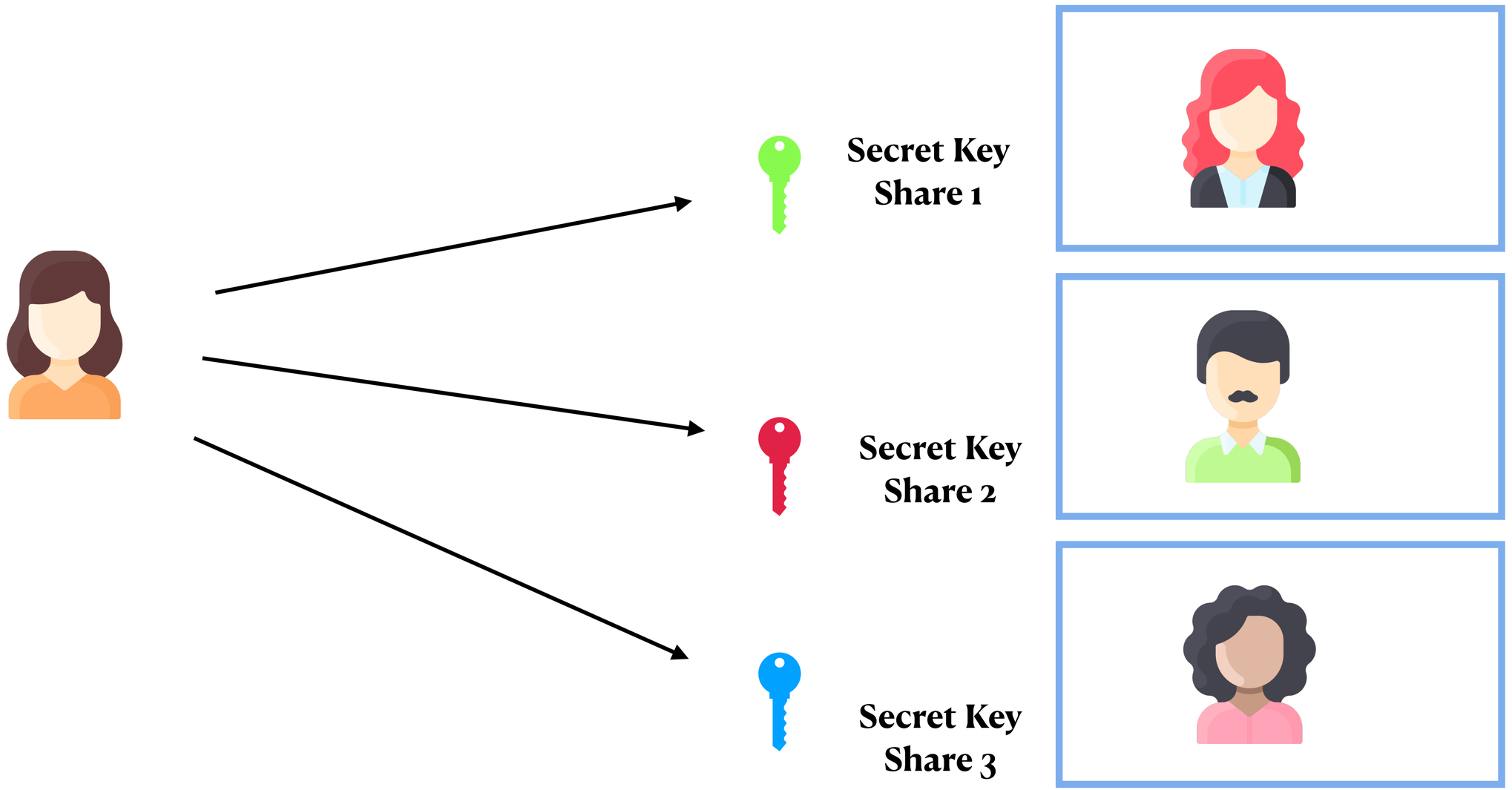
Chelsea Komlo

University of Waterloo,
NEAR One

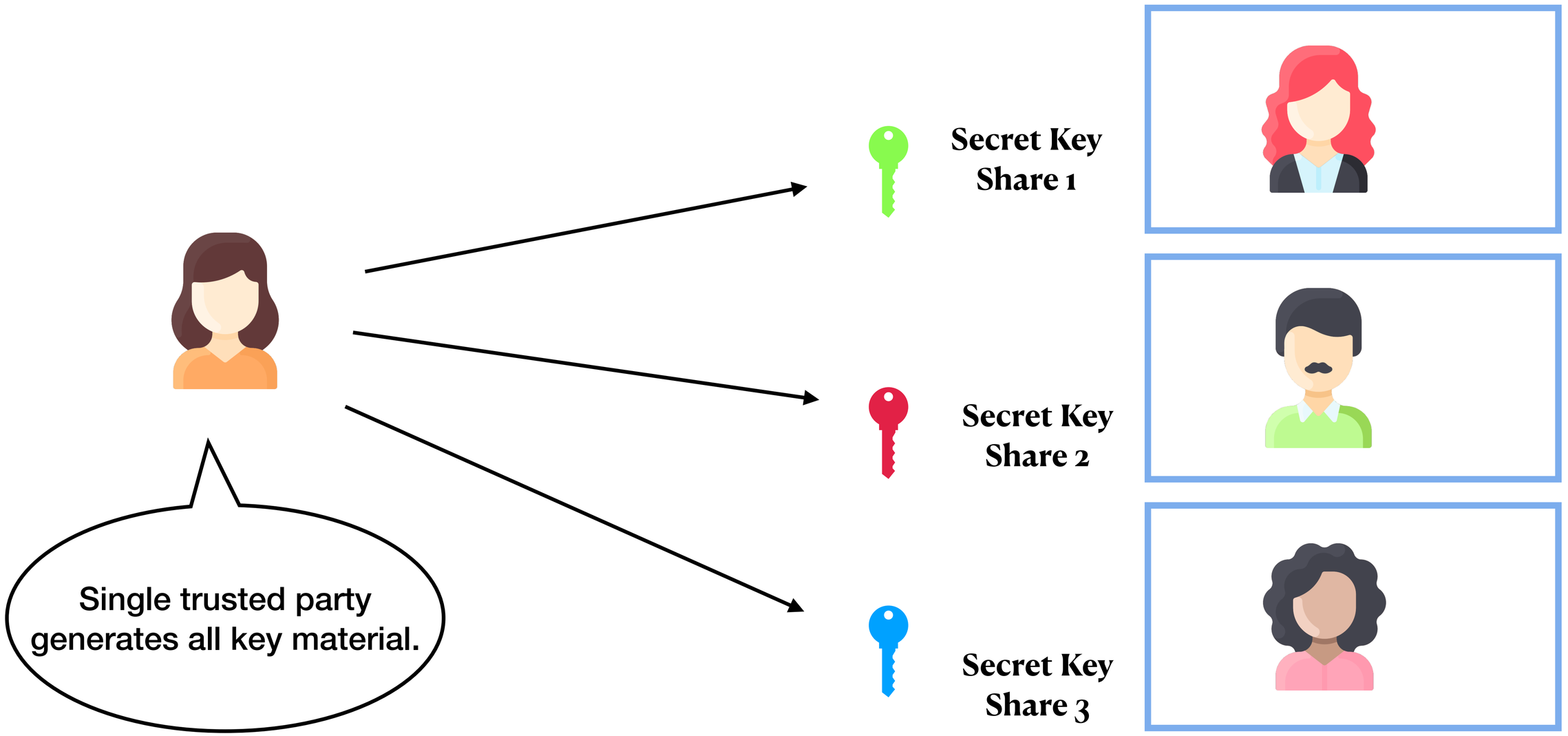
February 19, 2026

Motivation

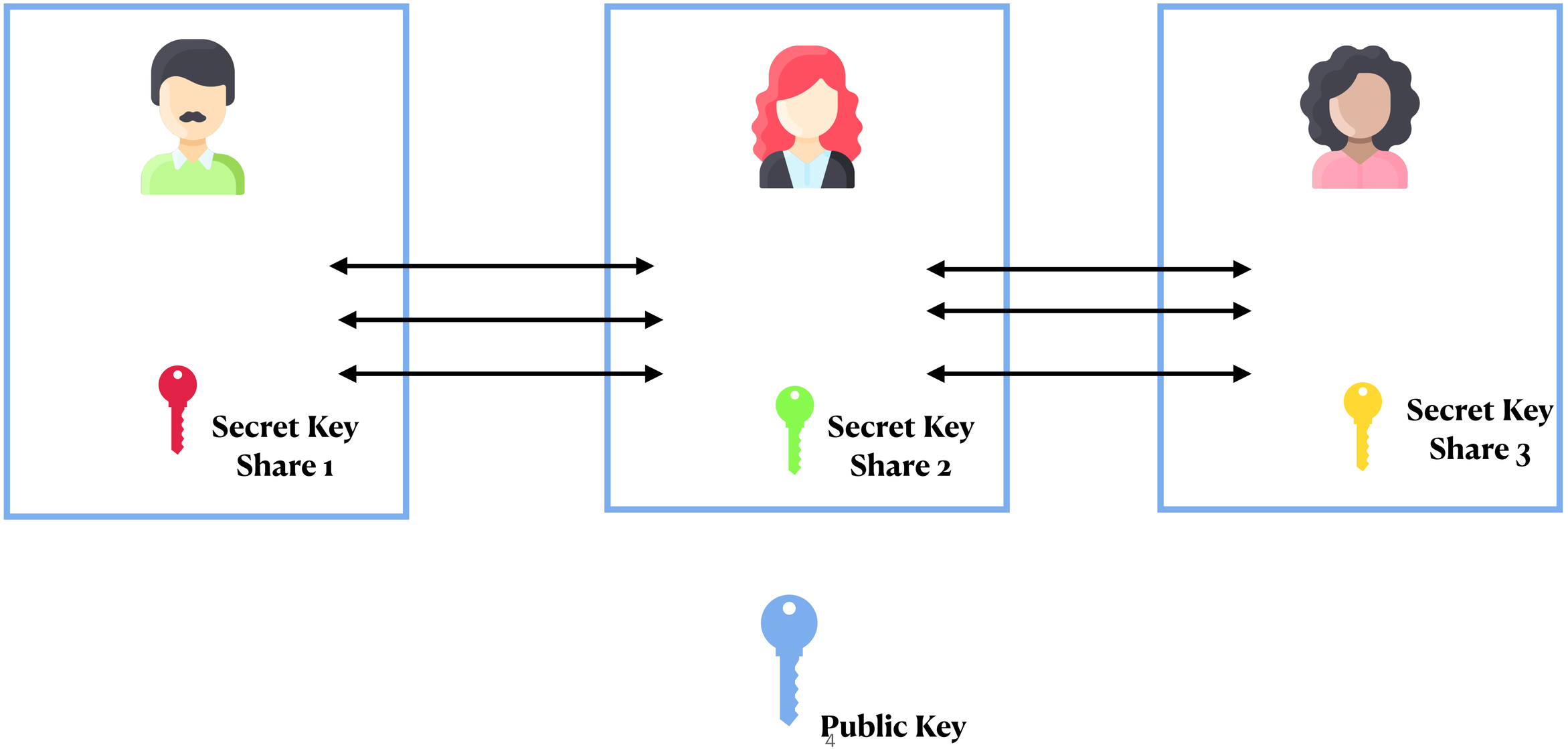
Threshold Key Generation, Trusted Dealer



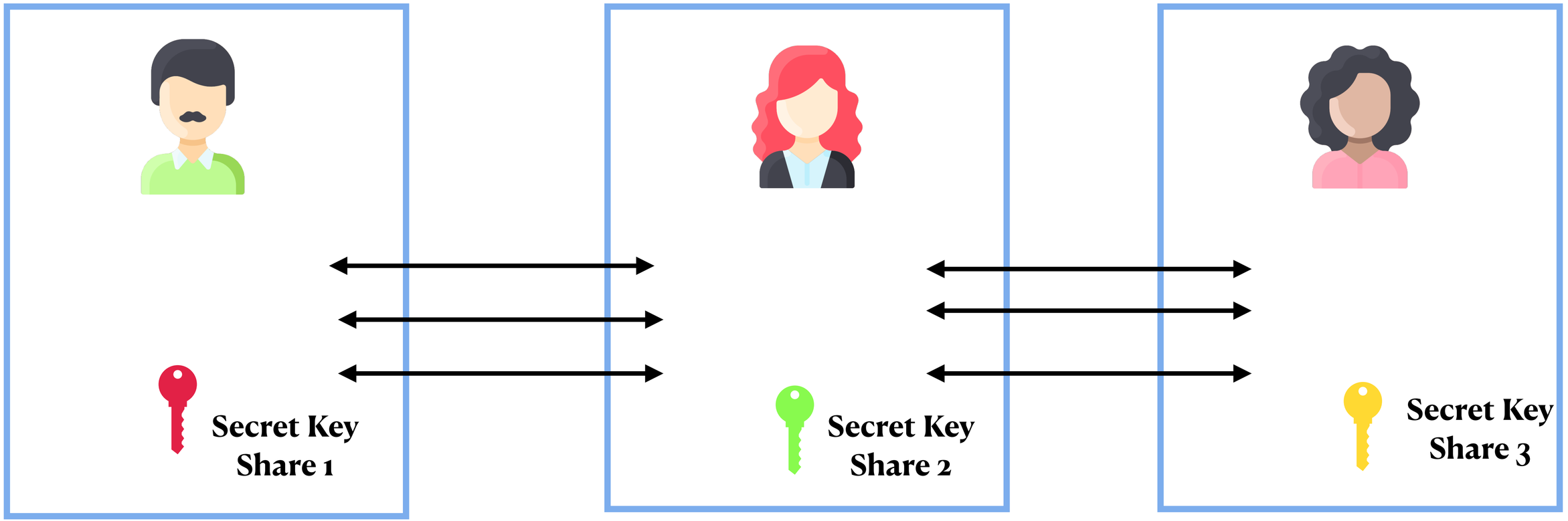
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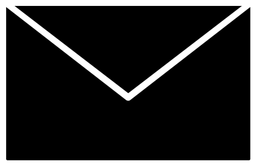
Distributed Key Generation



Distributed Key Generation



Uses for Distributed Key Generation



Threshold Signatures



Random Beacons



Threshold Encryption

Prior Work

Multi-Round

Secure Distributed Key Generation for Discrete-Log Based Cryptosystems

Rosario Gennaro*, Stanisław Jarecki†, Hugo Krawczyk‡and Tal Rabin*

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BIP:
Title: ChillDKG: Distributed Key Generation for FROST
Author: Tim Ruffing <crypto@timruffing.de>
        Jonas Nick <jonas@n-ck.net>
Status: Draft
License: CC0-1.0
License-Code: MIT
Type: Informational
Created:
Post-History:
Comments-URI:
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ChillDKG: Distributed Key Generation for FROST

Round-Optimal, Fully Secure Distributed Key Generation

Jonathan Katz*

Google and University of Maryland
jkatz@dfns.co, jkatz2@gmail.com

1-Round*

Non-interactive distributed key generation and key resharing

Jens Groth[1]

Non-interactive VSS using Class Groups and Application to DKG

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Exponent-VRFs and Their Applications

Dan Boneh[1], Iftach Haitner[2,3,⋆], Yehuda Lindell[4], and Gil Segev[4,5,⋆]

Prior Work

7 rounds with robustness

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1 online round; with preprocessing

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High bandwidth

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Additional assumptions / groups

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Additional assumptions / groups

Requires additional consensus round

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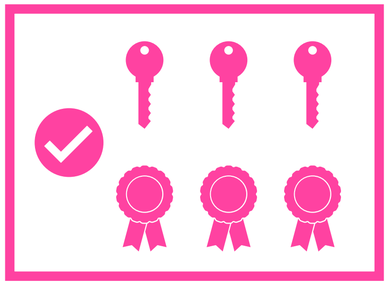
Dan Boneh[1], Iftach Haitner[2,3,⋆], Yehuda Lindell[4], and Gil Segev[4,5,⋆]

Groth (ElGamal) DKG



Each party plays the role of the dealer.







ElGamal Encrypted Secret Share Chunks

+

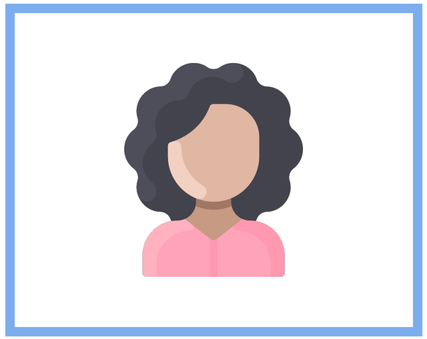
Zero-knowledge Proofs

+

Polynomial Commitment





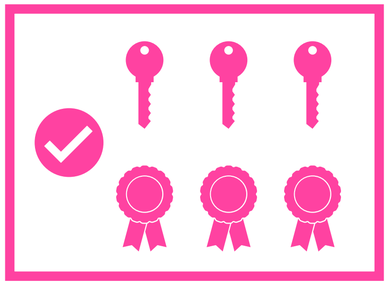


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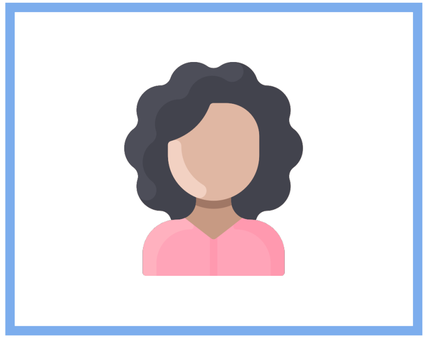
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Polynomial Commitment

Each Party:





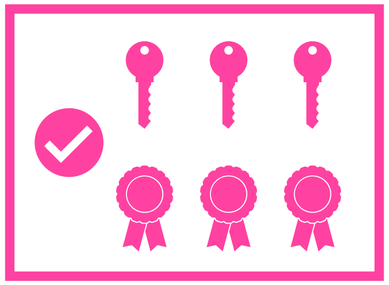


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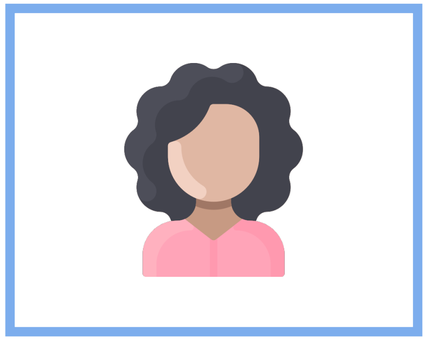
Zero-knowledge Proofs

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Polynomial Commitment







Each Party:

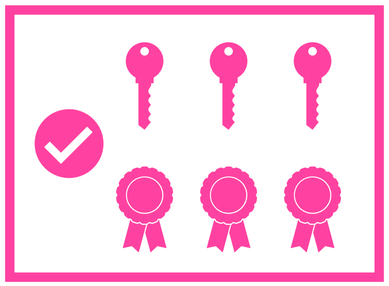
1. **Verifies proofs for all other parties**

Groth (ElGamal) DKG



Each party plays the role of the dealer.







ElGamal Encrypted Secret Share Chunks

+

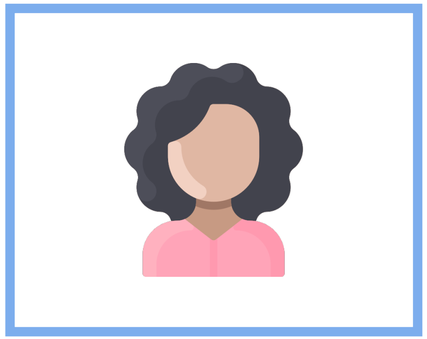
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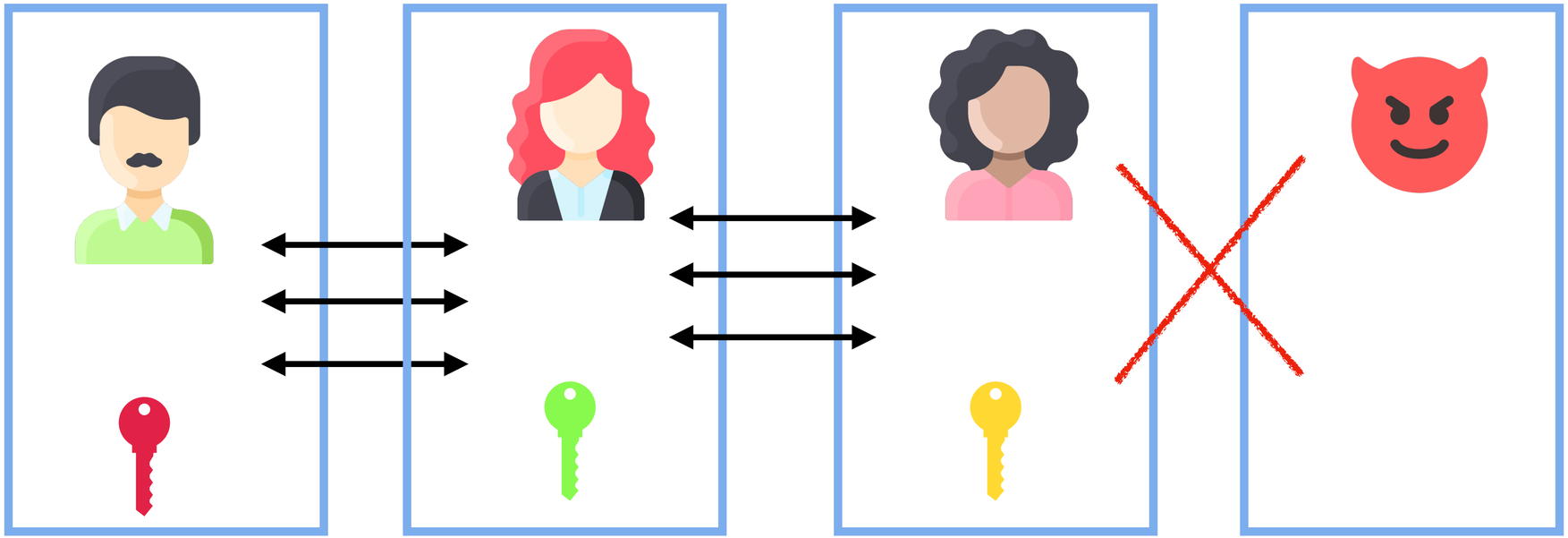


Each Party:

1. **Verifies proofs for all other parties**
2. **Elgamal decrypts each chunk**

Research Question: Can we design a more efficient one-round DKG?

Bonus Feature: Robustness

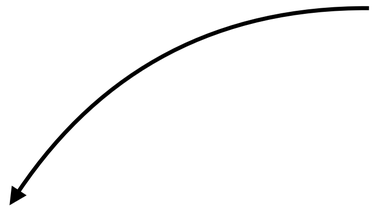




Public Key

Building Block: Two-Party eVRF

Exponent Verifiable Random Function (eVRF)



$$(r, R = g^r, \pi) \leftarrow \text{eVRF}.\text{Evaluate}(\text{sk}, m)$$

$$0/1 \leftarrow \text{eVRF}.\text{Verify}(\text{pk}, m, R, \pi)$$

Two-Party Exponent Verifiable Random Function (eVRF)

$$(r, R = g^r, \pi) \leftarrow \text{eVRF}.\text{Evaluate}(\text{sk}_1, \text{PK}_2, m)$$

$$0/1 \leftarrow \text{eVRF}.\text{Verify}(\text{pk}_1, \text{PK}_2, m, R, \pi)$$

Correctness of TeVRF: Symmetry

$$(r, R, \cdot) = \text{TeVRF . Evaluate}(\text{sk}_i, \text{pk}_j, m) = \text{TeVRF . Evaluate}(\text{sk}_j, \text{pk}_i, m)$$

$$0/1 \leftarrow \text{eVRF . Verify}(\text{pk}_i, \text{pk}_j, m, R, \pi)$$

Building Block:
Non-Interactive Key Exchange (NIKE)

$$S \leftarrow \mathsf{NIKE}.\mathsf{SharedKey}(\mathsf{sk}_i, \mathsf{pk}_j, m)$$

Relation

$$\mathcal{R}_{\mathsf{eVRF}^\dagger} = \Big\{ (\mathbb{G}_{\mathsf{in}} \subset E(\mathbb{F}_p); \{\mathsf{H}_{\mathbb{G}_{\mathsf{in}},j}(\mathsf{msg}), \mathsf{PK}_j \in \mathbb{G}_{\mathsf{in}}\}_{j=1}^{2}, R \in \mathbb{G}_{\mathsf{out}}, \beta \in \mathbb{F}_p) \ ; \ (\mathsf{sk}_1 \in \mathbb{F}_s) \Big\}$$

$$\begin{aligned}
\text{iff} \quad & (0) \quad \mathsf{PK}_1 \stackrel{?}{=} g_{\mathsf{in}}^{\mathsf{sk}_1} \in \mathbb{G}_{\mathsf{in}} \\
& (1) \quad S \leftarrow \mathsf{PK}_2^{\mathsf{sk}_1} \in \mathbb{G}_{\mathsf{in}} \\
& (2) \quad k_0 \leftarrow S.X \in \mathbb{F}_p \\
& (3) \quad k \leftarrow \mathsf{int}(k_0) \in \mathbb{Z} \\
& (4) \quad T_1 \leftarrow \mathsf{H}_{\mathbb{G}_{\mathsf{in}},1}(\mathsf{msg})^k \in \mathbb{G}_{\mathsf{in}} \\
& (5) \quad T_2 \leftarrow \mathsf{H}_{\mathbb{G}_{\mathsf{in}},2}(\mathsf{msg})^k \in \mathbb{G}_{\mathsf{in}} \\
& (6) \quad r_1 \leftarrow T_1.X \in \mathbb{F}_p \\
& (7) \quad r_2 \leftarrow T_2.X \in \mathbb{F}_p \\
& (8) \quad r \leftarrow \beta \cdot r_1 + r_2 \in \mathbb{F}_p \\
& (9) \quad R \stackrel{?}{=} g_{\mathsf{out}}^r \in \mathbb{G}_{\mathsf{out}}
\end{aligned}$$

Relation

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iff

(0) $\mathsf{PK}_1 \stackrel{?}{=} g_{\mathsf{in}}^{\mathsf{sk}_1} \in \mathbb{G}_{\mathsf{in}}$

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(9) $R \stackrel{?}{=} g_{\mathsf{out}}^{r} \in \mathbb{G}_{\mathsf{out}}$

DH Key exchange
Context: *msg*
Uniform mod p

Relation

$$\mathcal{R}_{\mathsf{eVRF}^\dagger} = \Big\{ (\mathbb{G}_{\mathsf{in}} \subset E(\mathbb{F}_p); \{\mathsf{H}_{\mathbb{G}_{\mathsf{in}},j}(\mathsf{msg}), \mathsf{PK}_j \in \mathbb{G}_{\mathsf{in}}\}_{j=1}^{2}, R \in \mathbb{G}_{\mathsf{out}}, \beta \in \mathbb{F}_p) \ ; \ (\mathsf{sk}_1 \in \mathbb{F}_s) \Big\}$$

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(9) $R \stackrel{?}{=} g_{\mathsf{out}}^{r} \in \mathbb{G}_{\mathsf{out}}$

DH Key exchange
Context: *msg*
Uniform mod p

Relation

$$\mathcal{R}_{\mathsf{eVRF}^\dagger} = \left\{ (\mathbb{G}_{\mathsf{in}} \subset E(\mathbb{F}_p); \{\mathsf{H}_{\mathbb{G}_{\mathsf{in}},j}(\mathsf{msg}), \mathsf{PK}_j \in \mathbb{G}_{\mathsf{in}}\}_{j=1}^{2}, R \in \mathbb{G}_{\mathsf{out}}, \beta \in \mathbb{F}_p) \ ; \ (\mathsf{sk}_1 \in \mathbb{F}_s) \right\}$$

iff

(0) $\mathsf{PK}_1 \stackrel{?}{=} g_{\mathsf{in}}^{\mathsf{sk}_1} \in \mathbb{G}_{\mathsf{in}}$

(1) $S \leftarrow \mathsf{PK}_2^{\mathsf{sk}_1} \in \mathbb{G}_{\mathsf{in}}$

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(3) $k \leftarrow \mathsf{int}(k_0) \in \mathbb{Z}$

(4) $T_1 \leftarrow \mathsf{H}_{\mathbb{G}_{\mathsf{in}},1}(\mathsf{msg})^k \in \mathbb{G}_{\mathsf{in}}$

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Fixed-base scalar multiplications
770 constraints each
(Can be made even cheaper)

DH Key exchange
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Fixed-base scalar multiplications
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DH Key exchange
Context: *msg*
Uniform mod p

Free in Bulletproofs

Relation

$$\mathcal{R}_{\mathsf{eVRF}^\dagger} = \left\{ (\mathbb{G}_{\mathsf{in}} \subset E(\mathbb{F}_p); \{\mathsf{H}_{\mathbb{G}_{\mathsf{in}},j}(\mathsf{msg}), \mathsf{PK}_j \in \mathbb{G}_{\mathsf{in}}\}_{j=1}^{2}, R \in \mathbb{G}_{\mathsf{out}}, \beta \in \mathbb{F}_p) \ ; \ (\mathsf{sk}_1 \in \mathbb{F}_s) \right\}$$

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Fixed-base scalar multiplications
770 constraints each
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Free in Bulletproofs

DH Key exchange
Context: *msg*
Uniform mod p

Golden

Features

Features

- One-Round

Features

- One-Round
- Robust

Features

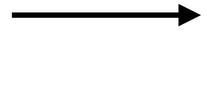
- One-Round
- Robust
- Efficient

Features

- One-Round
- Robust
- Efficient
- Composable (UC-Security)

Golden: Intuition











Each party plays the role of the dealer.

One-Time-Pad (NIKE) Encrypted Secret Shares

+

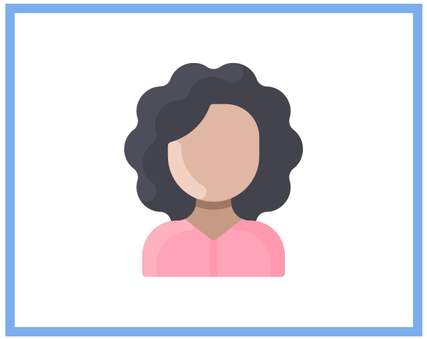
ZK Proof

+

Polynomial Commitment







Golden: Intuition



Each party plays the role of the dealer.







One-Time-Pad (NIKE) Encrypted Secret Shares

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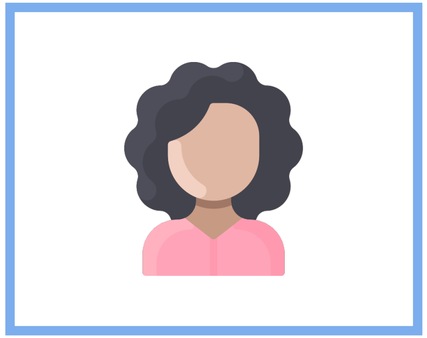
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Golden: Intuition



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One-Time-Pad (NIKE) Encrypted Secret Shares

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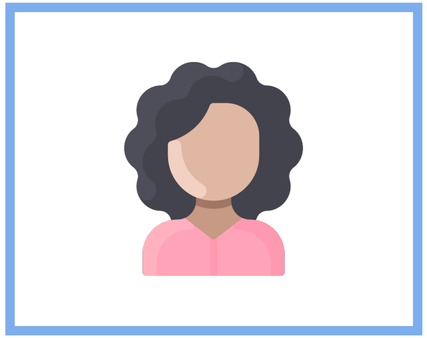
ZK Proof

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Polynomial Commitment







Each Party:

1. **Verifies ZK proofs for all other parties**

Golden: Intuition



Each party plays the role of the dealer.







One-Time-Pad (NIKE) Encrypted Secret Shares

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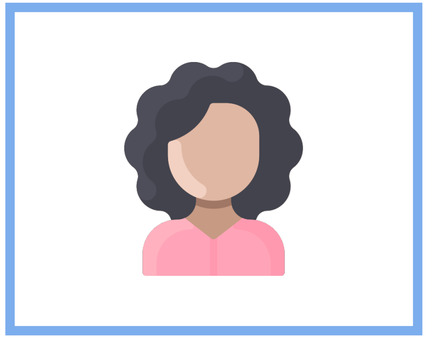
ZK Proof

+

Polynomial Commitment







Each Party:

1. **Verifies ZK proofs for all other parties**
2. **Derives the NIKE shared secret for each party**

Golden: Intuition



Each party plays the role of the dealer.







One-Time-Pad (NIKE) Encrypted Secret Shares

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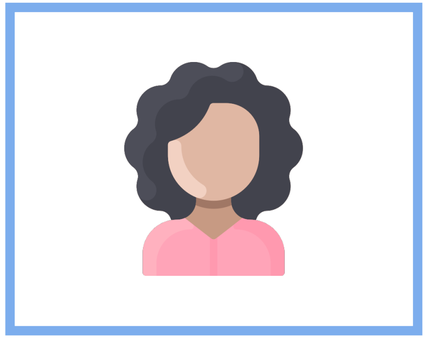
ZK Proof

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Polynomial Commitment







Each Party:

1. **Verifies ZK proofs for all other parties**
2. **Derives the NIKE shared secret for each party**
3. **(Linearly) decrypts their shares**

Construction

PK_{ID}^{1}





PK_{ID}^{2}



PK_{ID}^{n}

Combine

$$PK = \prod_{i=1}^{n} A_{i,0}$$

$$sk_i = \sum_{j=1}^{n} f_j(i)$$

Construction

$$\omega_i \leftarrow \mathbb{Z}_q$$

$\mathsf{PK}_{\mathsf{ID}}^{1}$





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$\mathsf{PK}^1_{\mathsf{ID}}$

$$\{f_i(j)\}_{j=1}^{n}, (A_{i,k})_{k=0}^{t-1} \leftarrow \text{Feldman . Share}(\omega_i, n, t)$$





$\mathsf{PK}^2_{\mathsf{ID}}$



PK^n_{ID}

Combine

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Construction

Shamir secret shares

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Construction

Shamir secret shares

Polynomial commitment to f

$\omega_i \leftarrow \mathbb{Z}_q$

$\mathsf{PK}_{\mathsf{ID}}^1$ $\{f_i(j)\}_{j=1}^n, (A_{i,k})_{k=0}^{t-1} \leftarrow \text{Feldman . Share}(\omega_i, n, t)$





$\mathsf{PK}_{\mathsf{ID}}^2$



PK_{ID}^n

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$\mathsf{PK}^1_{\mathsf{ID}}$



$\{f_i(j)\}_{j=1}^{n}, (A_{i,k})_{k=0}^{t-1} \leftarrow \mathsf{Feldman.Share}(\omega_i, n, t)$

$r_{i,j}, R_{i,j}, \pi_{i,j} \leftarrow \mathsf{TeVRF.Evaluate}(\mathsf{sk}^1_{\mathsf{ID,i}}, \mathsf{PK}^1_{\mathsf{ID,j}}, \mathsf{msg}_i)$



$\mathsf{PK}^2_{\mathsf{ID}}$



PK^n_{ID}

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Polynomial commitment to f

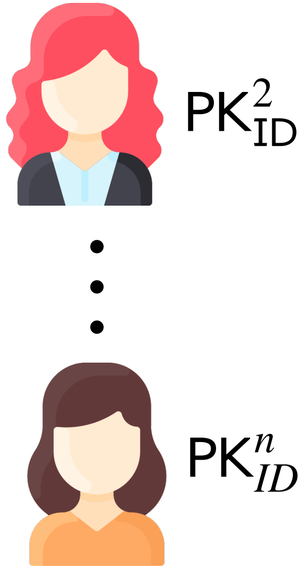
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Do for all parties



Combine

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$$\mathsf{sk}_i = \sum_{j=1}^{n} f_j(i)$$

Construction

Shamir secret shares

Polynomial commitment to f

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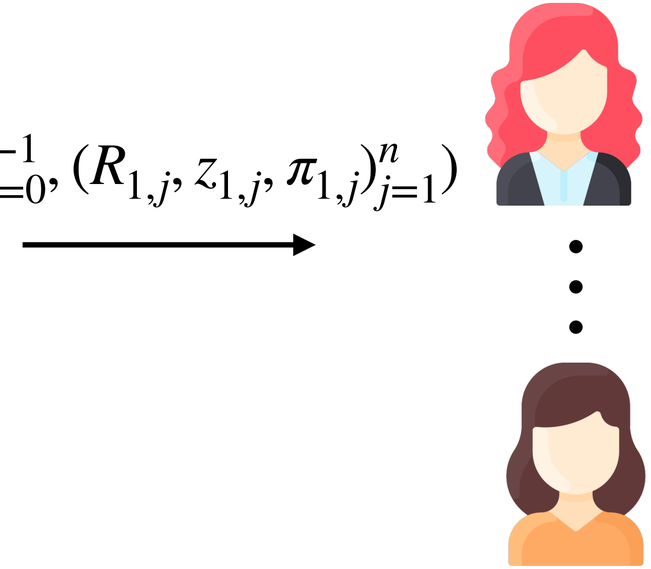


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$\mathsf{PK}^2_{\mathsf{ID}}$

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PK^n_{ID}

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return $\perp$ if $\mathsf{TeVRF.Verify}(\mathsf{PK}^1_{\mathsf{ID,i}}, \mathsf{PK}^1_{\mathsf{ID,j}}, \mathsf{msg}_i, R_{i,j}, \pi_{i,j}) \neq 1$



$\mathsf{PK}^2_{\mathsf{ID}}$

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$$X_{j,i} = \prod_{k=0}^{t-1} A_{i,k}^{j^k} = g^{f_j(i)}$$



$\mathsf{PK}^2_{\mathsf{ID}}$

$((A_{i,k})_{k=0}^{t-1}, (R_{1,j}, z_{1,j}, \pi_{1,j})_{j=1}^{n})$

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$$((A_{i,k})_{k=0}^{t-1}, (R_{1,j}, z_{1,j}, \pi_{1,j})_{j=1}^{n})$$

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return $\perp$ if $\mathsf{TeVRF.Verify}(\mathsf{PK}^1_{\mathsf{ID,i}}, \mathsf{PK}^1_{\mathsf{ID,j}}, \mathsf{msg}_i, R_{i,j}, \pi_{i,j}) \neq 1$

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Combine

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Construction

Shamir secret shares

Polynomial commitment to f

$\omega_i \leftarrow \mathbb{Z}_q$

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Derive n shared secrets

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Decrypt!

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Ideal Functionality F_{DKG}^{B}

[KMSSSY26]

Adversary

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t-1 parties

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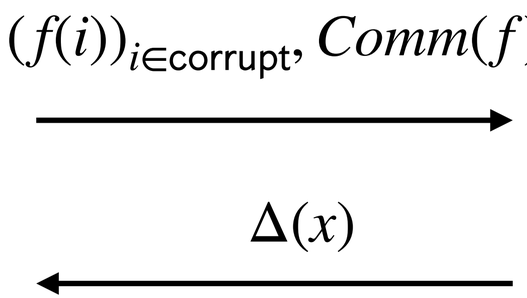
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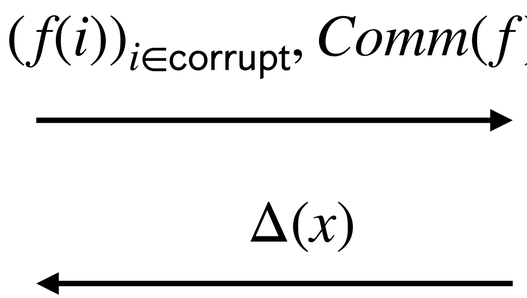


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Adversary can bias keys, but by a imited amount, with limited information.

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Performance

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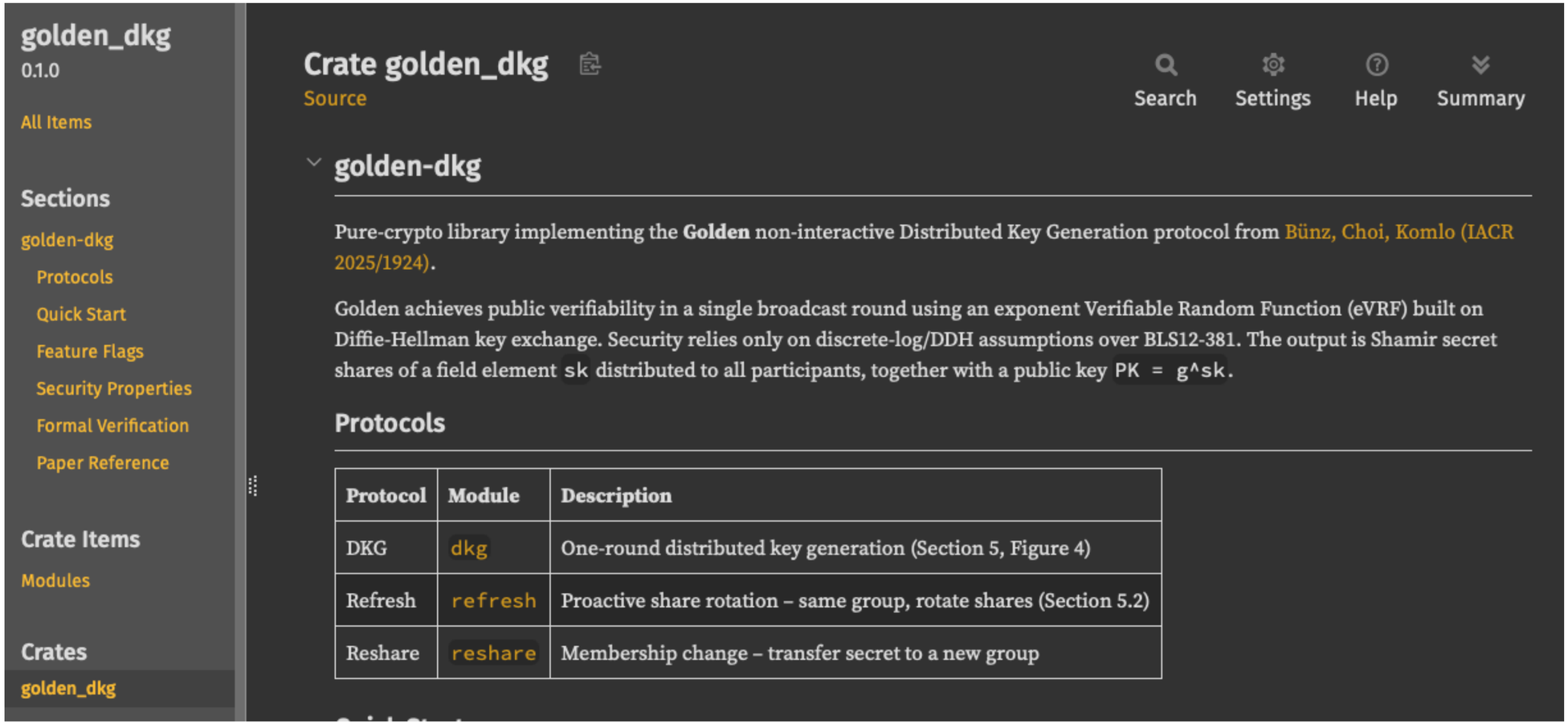
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- In Golden: each node
  - proofs n EVRFs (batch generation)
  - checks n proofs each for n VRFs (batch verification)

	Golden (optimized)		Groth DKG [48]	
Participants	Communication	Runtime	Communication	Runtime
2	1.7 kb	0.4 s	59.4 kb	3.9 s
10	22 kb	2.4 s	1.3 MB	5.6 s
50	223 kb	13.5 s	27.8 MB	40.5 s
100	699 kb	35.8 s	108.6 MB	130.8 s

Scheme	Rounds	Bandwidth	Computation	Assumption
Fouque & Stern [42]*	1	$\mathcal{O}(n^2t)$	$\mathcal{O}(nt)$	Paillier
ElGamal-DKG/Groth [48]	1	$\mathcal{O}(n^2tc\ell)$	$\mathcal{O}(ntc\ell)$	DL/DDH
Cascudo & David [19] Kate et al. [53]	1	$\mathcal{O}(n^2t)$	$\mathcal{O}(nt)$	Class Groups
Katz [54]	3 (2 preprocessing)	$\mathcal{O}(n^2t)$	$\mathcal{O}(\binom{n}{t})$	DL
BHLS25-DKG [12]	2†	$\mathcal{O}(n^2t)$	$\mathcal{O}(nt)$	DL/DDH
Golden (this work)	1	$\mathcal{O}(n^2t)$	$\mathcal{O}(nt)$	DL/DDH

Use in Practice!



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Thank you!

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$$(i, sk_i), PK, \{PK_i\}_{i=1}^n \longrightarrow$$

Recovering sk

$$sk \leftarrow \sum_{i \in S} sk_i \lambda_i,\ S \subseteq \{1, \ldots, n\}, |S| \geq t$$

- t-out-of-n
- λ_i is the ith Lagrange coefficient; determined using only (i, S)
- t points uniquely define a polynomial of degree $t - 1$

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Where Y_0, Y_i are uniformly random in the group.